

Shell Scripting-2 (cntd..)

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Operators

- Arithmetic Operators
- Relational Operators
- Boolean Operators
- String Operators
- File Test Operators

Bourne shell originally didn't ve any mechanism for arithmetic operations

- Bourne shell uses external programs like **awk** or **expr** to perform arithmetic operations

example

```
#!/bin/sh  
val= `expr 2 + 2`  
echo "total value: $val"
```

Note:-

1. There must be **spaces** between operators and expressions 2 + 2 is correct, not 2+2
2. The complete expression must be enclosed between ` `, called **backtick**

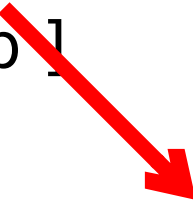
\$a=10 and \$b=20

operator

- +
- -
- *
- /
- %
- ==
- !=

example

- `expr \$a + \$b`
- `expr \$a - \$b`
- `expr \$a \* \$b`
- `expr \$a / \$b`
- `expr \$a % \$b`
- [\$a == \$b]
- [\$a != \$b]



Space
s

Relational Operators

Operator

- -eq
- -ne
- -gt
- -lt
- -ge
- -le

Example

- [\$a -eq \$b]
- [\$a -ne \$b]
- [\$a -gt \$b]
- [\$a -lt \$b]
- [\$a -ge \$b]
- [\$a -le \$b]

Boolean Operators

Operator

- !
- -o
- -a

Example

- [! false]
- [\$a -lt 20 -o \$b -gt 100]
- [\$a -lt 20 -a \$b -gt 100]

String Operators

a holds “abc” , b holds “efg”

Operator

- =
- !=
- -z
- -n

Example

- [\$a = \$b] <equal or not>
- [\$a != \$b]
- [-z \$a] <length is zero>
- [-n \$a] <length is non zero>

If statement (if then, if then else, if
elif else)

```
#!/bin/bash
count =10
if [ $count -eq 9 ]
then
    echo "cond is true"
else
    echo "cond is false"
fi
```

Example with string variable...

```
#!/bin/bash
```

```
word=abc
```

```
if [ $word == "abc" ]
```

```
then
```

```
    echo "cond is true"
```

```
fi
```

```
#!/bin/bash    cntd..  
word=a  
if [[ $word == "b" ]]  
then  
    echo "cond b is true"  
elif [[ $word=="a" ]]  
then  
    echo "cond a is true"  
else  
    echo "cond is false"  
fi
```

Control Structures

```
if expression  
then  
    command1  
else  
    command2  
fi
```

if.....else statements

- if.....fi
- if.....else.....fi
- if.....elif.....else.....fi